

LAKE DISTRICT



Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines structurally insulated panel construction for repeatable, high-performance row houses; passive solar design, natural ventilation, FSC-certified timber and low-VOC finishes; and efficient water fixtures and landscaping, with more than half of construction waste targeted for diversion from landfill. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Miroslava, Iași County, Romania
Building use / function	Large residential neighbourhood of row houses
Project type (new build, refurbishment, extension)	New build; phased masterplan

Plot area / Floor area	More than 600 homes planned; exact certified floor area not publicly disclosed
Year / phase	Phased development / certification process described in the RoGBC toolkit
Climate zone	Temperate continental
Thematic focus / innovation	Structurally insulated panels; passive solar orientation; natural ventilation; FSC wood; low-VOC finishes; construction-waste diversion
Investor / building owner	American Eco Homes SRL
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; timber/SIP suppliers; landscape and waste-management partners
Contact person	American Eco Homes / RoGBC
Further information	https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Lake District is a new build; phased masterplan project in Miroslava, Iași County, Romania. It functions as large residential neighbourhood of row houses. Its main best-practice contribution is A large-scale attempt to industrialise and replicate green-home specifications across more than 600 planned dwellings. The documented strategy brings together structurally insulated panel construction for repeatable, high-performance row houses, passive solar design, natural ventilation, FSC-certified timber and low-VOC finishes, and efficient water fixtures and landscaping, with more than half of construction waste targeted for diversion from landfill. The toolkit describes planned measures and certification progress; public completion and measured-performance data were not identified. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - Structurally insulated panel construction for repeatable, high-performance row houses.
2. Integrated systems and operation - Passive solar design, natural ventilation, fsc-certified timber and low-voc finishes.
3. Wider value - Efficient water fixtures and landscaping, with more than half of construction waste targeted for diversion from landfill.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.